# CALCULUS WITH

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TOPIC TWO --- ANSWER KEY | TOPIC SERIES

## Limits

Compiled and taught by Asante

MATH 152 | Calculus & Analysis | KNUST BME1

# Problem Set --- Worked Solutions

1. $\lim_{x\to 9} \dfrac{\sqrt{x}-3}{x-9}$ **Correct answer: (B)**
   Direct substitution gives $\frac{0}{0}$. Multiply by the conjugate $\sqrt{x}+3$:

   $$\frac{\sqrt{x}-3}{x-9}\cdot\frac{\sqrt{x}+3}{\sqrt{x}+3}=\frac{x-9}{(x-9)(\sqrt{x}+3)}=\frac{1}{\sqrt{x}+3}$$

   So the limit is $\dfrac{1}{\sqrt{9}+3}=\dfrac{1}{6}$.

2. $\lim_{x\to\infty} \dfrac{4x^3-x}{2x^3+7}$ **Correct answer: (A)**
   Divide top and bottom by $x^3$:

   $$\lim_{x\to\infty}\frac{4-\frac{1}{x^2}}{2+\frac{7}{x^3}}=\frac{4}{2}=2$$

3. $\lim_{x\to 0} \dfrac{\sin(3x)}{x}$ **Correct answer: (C)**
   Rewrite as $3\cdot\dfrac{\sin(3x)}{3x}$. As $x\to 0$, $3x\to 0$ too, so $\dfrac{\sin(3x)}{3x}\to 1$ by the foundational trig limit. The answer is $3\cdot 1=3$.

4. $\lim_{x\to 0} \dfrac{|x|}{x^2}$ **Correct answer: (C)**
   For $x\neq 0$, $\dfrac{|x|}{x^2}=\dfrac{1}{|x|}$, which grows without bound as $x\to 0$ from either side. The limit does not exist — it diverges to $+\infty$ from both sides (so unlike Example 5 in the booklet, the one-sided limits actually agree here, but neither is finite).

5. **T** $\lim_{x\to 1} \ln x$ **Correct answer: (B)**
   $\ln x$ is continuous at $x=1$, so direct substitution works: $\ln 1=0$.

6. $\lim_{x\to 0} x^4\cos\left(\dfrac{1}{x}\right)$ **Correct answer: (A)**
   Since $-1\leq\cos(1/x)\leq 1$ for all $x\neq 0$, multiplying by $x^4\geq 0$ gives $-x^4\leq x^4\cos(1/x)\leq x^4$. Both bounds $\to 0$ as $x\to 0$, so by the Squeeze Theorem the limit is 0.

7. Piecewise limit at $x=3$ for $h(x)=\begin{cases}2x & x\leq 3\\ x+4 & x>3\end{cases}$ **Correct answer: (C)**

8. $\lim_{x \to -1^+} \sin^{-1} x$ **Correct answer: (B)**
$\sin^{-1} x$ is continuous on $[-1, 1]$, so direct substitution from within the domain works: $\sin^{-1}(-1) = -\frac{\pi}{2}$.

9. Root of $f(x) = \cos x - x$ between 0 and 1 **Correct answer: (A)**
$f$ is continuous everywhere (difference of continuous functions). Check the endpoints:

$$f(0) = \cos(0) - 0 = 1 \qquad f(1) = \cos(1) - 1 \approx -0.46$$

Since $f(0) > 0 > f(1)$, by the IVT there exists some $c \in (0, 1)$ with $f(c) = 0$.

10. **T** Is $f(x) = \sqrt[3]{x}$ continuous and differentiable at $x = 0$? **Correct answer: (B)**
*Continuous:* yes — $f(0) = 0$ and $\lim_{x \to 0} \sqrt[3]{x} = 0$ from both sides.
*Differentiable:* no — the slope $\dfrac{f(h) - f(0)}{h} = \dfrac{h^{1/3}}{h} = h^{-2/3}$ blows up to $+\infty$ as $h \to 0$ from either side. The graph has a vertical tangent at the origin, so no finite derivative exists there — a second flavour of "continuous but not differentiable," different from the sharp corner in $|x|$.